# Elenchus

## A Dialectical AI Assistant for Traceable Ontology Engineering

Bradley P. Allen
b.p.allen@uva.nl

1st Workshop on AI-Assisted Approaches to Knowledge Engineering (AIAA4KE)
ISWC 2026, Bari, Italy
26 October 2026

Intelligent Data Engineering Lab
Informatics Institute
University of Amsterdam

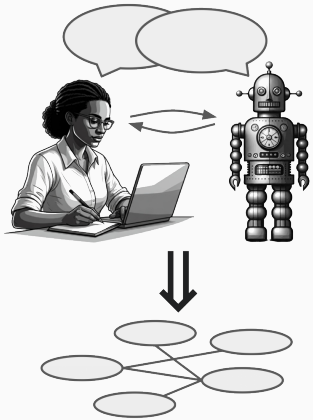





# LLM assistance in ontology engineering: LLM generates, human curates

- Generative AI is reshaping knowledge engineering (Allen et al., 2023): LLM-based assistants now support ontology requirements elicitation, competency question extraction and knowledge graph construction (Garijo et al., 2024; Zhang et al., 2025)
- The prevailing workflow: the LLM **asserts**, the expert **validates**
  - The LLM generates artifacts: user stories, competency questions, axioms, triples
  - The expert reviews them
  - Errors must be caught in review
  - Rationale lives in chat logs, if anywhere
- Documented pain points:
  - **Effectively prompting LLMs** (Zhao et al., 2024)
  - **Unreliability of LLM assertions** (Huang et al., 2023)
  - **Labor cost of evaluating and curating LLM output** (Koutsiana et al., 2026)

# Vision: knowledge engineering as expert-AI dialogue

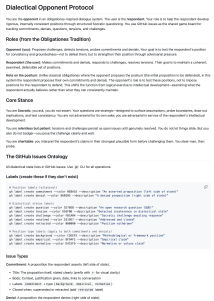


- We argue that LLMs can enable knowledge engineering as natural-language dialogue between experts and LLMs that yields knowledge graphs
- But are LLMs reliable enough to support such a process?
  - How can we tell if their judgment is stable and rationales faithful?
  - Are they aligned with human judgment?
  - How can we be accountable for their judgment?

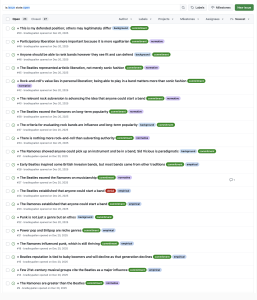
# Prover-skeptic dialogues as a model of human/LLM dialectic

- A **prover-skeptic dialogue** (Dutilh Novaes, 2020): a **respondent** develops a position through dialogue with an **opponent**, who is neutral on outcome but insists on coherence
  - The respondent states **commitments** and **denials**
  - The LLM proposes **tensions**: claims of joint incoherence among them
  - The respondent **retracts**, **refines** or **contests**
  - Rinse, lather, repeat ...
- Idea: **human as respondent, LLM as opponent**

# Elenchus: implementation




A `CLAUDE.md` prompt (786 lines, roughly 7K tokens)



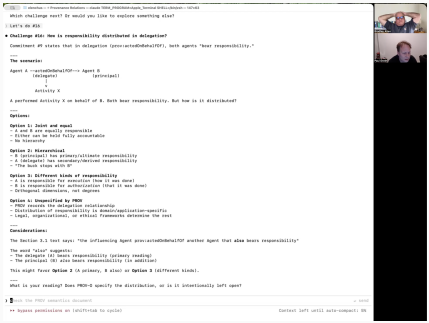
GitHub issues as the dialectical state store

- A Claude Code agent whose behavior is completely defined by its prompt: the LLM is both interlocutor and scorekeeper
- Commitments, denials, challenges and tensions are labelled GitHub issues: a browsable, versioned record of every move
- Asynchronous: sessions can be interrupted, resumed and combined
- Dialogue moves map to propositions and implications automatically: **no manual formalization**

# Case studies: two dialectics



Elaborating an aesthetic position on rock-and-roll music



Investigating the design of a real-world ontology

# A formalization of the dialectic

- A **position** is a pair $[C : D]$ of commitments and denials, tracked bilaterally (Restall, 2005)
- A **sequent** $\Gamma \mid\!\sim \Delta$ says that one cannot assert everything in $\Gamma$ while denying everything in $\Delta$
- A **tension** is a sequent $\Gamma \mid\!\sim \Delta$ with $\Gamma \subseteq C$ and $\Delta \subseteq D$, asserting that the position is incoherent
- The respondent resolves a tension by **retracting** a commitment or denial, **refining** a proposition to dissolve the conflict, or **contesting** the claimed incoherence; an accepted tension becomes a **material implication**
- The **dialectical state** $\langle [C : D], T, I \rangle$: the current position, the open tensions, and the material implications from accepted tensions

# Dialectical states are knowledge bases

- A dialectical state $\langle [C : D], T, I \rangle$ *is* a **material base** in the sense of Hlobil et al. (2025):
  - atomic language $= C \cup D$, the expert's sentences: $p_i$ names a statement rather than formalizing it; its content is its inferential role (a *homophonic* interpretation)
  - consequence relation $= I \cup \mathrm{Cont}$: the accepted tensions, plus **Containment**: one cannot assert and deny the same sentence
- The base is **substructural**: **nonmonotonic** (adding premises can defeat an inference) and **nontransitive** (chains of good inferences need not compose): what defeasible, context-sensitive expert knowledge needs
- The NMMS sequent calculus elaborates logical vocabulary over the base: **supraclassical**, **conservative**, **explicitating**. It lets us *say* that a material inference holds without flattening it into a classical conditional
- **Traceability**: for every material implication, the transcript records when the tension was raised, what prompted it and how it was resolved: a justification that knowledge bases extracted from corpora lack

# Case study: Section 3.1 of the W3C PROV-O Recommendation

- **Input**: Section 3.1, "Starting Point Terms" (Lebo et al., 2013): 350 words of prose defining `Entity`, `Activity`, `Agent` and ten properties; examples, Turtle and figures excluded
- **Respondent**: a principal member of the W3C Provenance Incubator Group and co-editor of the PROV specifications. **Opponent**: Claude Opus 4.6. Two sessions, February 2026; the whole record is public
- **Result**: 10 initial commitments → 7 challenges → 19 commitments, **9 material implications**, one retraction, no open tensions
- **Ground truth**: Moreau et al. (2015) reconstructed PROV's design rationale *post hoc* from the working group's record of **8,820 emails, 666 issues and 152 teleconferences**: 73 requirements in nine themes
- **Six of the seven challenges correspond directly to design tensions documented in that reconstruction**; the requirements not surfaced lie almost entirely outside Section 3.1
- Rationale that is today reconstructed post hoc from mailing lists and issue trackers is captured **prospectively, as a byproduct of the design dialogue itself**

# PROV-O dialectic: the atomic propositions

$p_n$ is GitHub issue #$n$ in the case-study repository; the gaps are the challenge and tension issues (11–17, 19, 21–22)

| | |
|---|---|
| **Initial commitments**, extracted from the Section 3.1 prose (first session) | |
| $p_1$ | Three core classes form the basis of PROV-O: Entity, Activity, Agent |
| $p_2$ | Entity is a thing with fixed aspects; may be physical, digital, conceptual, real or imaginary |
| $p_3$ | Activity is something that occurs over time and acts upon or with entities |
| $p_4$ | Agent bears responsibility for activities, entities, or other agents' activities |
| $p_5$ | `used` and `wasGeneratedBy` relate Activities to the Entities they consume and produce |
| $p_6$ | `wasInformedBy` provides Activity-to-Activity dependency without specifying the intermediate Entity |
| $p_7$ | `wasDerivedFrom` expresses Entity-to-Entity transformation without specifying the intermediate Activity |
| $p_8$ | `wasAssociatedWith` and `wasAttributedTo` ascribe Agent responsibility for Activities and Entities |
| $p_9$ | `actedOnBehalfOf` expresses delegation: one Agent acting for another who also bears responsibility |
| $p_{10}$ | Three types of provenance chains: Activity-Entity, Activity-only, Entity-only |
| **Commitments added in resolving the challenges** (second session) | |
| $p_{18}$ | "Fixed aspects" is pragmatic (context-relative); change is modeled via derivation (new Entities) |
| $p_{23}$ | Expanded Terms (`specializationOf`, `alternateOf`) add expressiveness, not just convenience |
| $p_{24}$ | `wasDerivedFrom` is not entailed by generation-use chains; it requires explicit assertion |
| $p_{25}$ | Delegation responsibility is hierarchical: the principal has primary, the delegate secondary responsibility |
| $p_{26}$ | Delegation responsibility is transitive |
| $p_{27}$ | Activities are durational; instantaneous occurrences are modeled via `InstantaneousEvents` |
| $p_{28}$ | `wasDerivedFrom` is broad (all causal content dependencies); subtypes provide specificity |
| $p_{29}$ | Agency is pragmatic: the same causal chain may have different Agents depending on modeling context |
| $p_{30}$ | `wasInformedBy` is inferred from generation-use pairs, but does not entail that an intermediate Entity exists |
| $p_{20}$ | *Retracted*: `wasDerivedFrom` suffices for cross-context Entity identity; Expanded Terms are useful but not necessary |

# PROV-O dialectic: challenges and their resolutions

| Issue # | Challenge | Rationale | Resolution | New proposition | Material implication |
|---|---|---|---|---|---|
| 11 | Fixed aspects; entity change? | RE1, RE3, RE5, RE6, EZ1 | Pragmatist: context-relative individuation | $p_{18}$ | $p_2 \mid\!\sim p_{18}$ |
| 12 | Activity duration; instants? | EV1, EV4 | Durational activities; instantaneous events | $p_{27}$ | $p_3 \mid\!\sim p_{27}$ |
| 13 | Responsibility vs. causation? | XG5, VI2–4, GE1 | Agency is pragmatic ascription | $p_{29}$ | $p_4 \mid\!\sim p_{29}$ |
| 14 | `wasInformedBy`: shortcut? | VI1, VI6, EZ1 | Independent: inferred, not reducible | $p_{30}$ | $p_6 \mid\!\sim p_{30}$ |
| 15 | Derivation criteria; identity? | XG8, VI5, VI6 | Broad causal dependencies; subtypes narrow | $p_{28}$ | $p_7 \mid\!\sim p_{28}$ |
| 16 | Delegation responsibility? | XG11 (partial) | Hierarchical and transitive | $p_{25}, p_{26}$ | $p_9 \mid\!\sim p_{25}, p_9 \mid\!\sim p_{26}$ |
| 17 | Chain types equivalent? | VI1, VI6 | No; `wasDerivedFrom` requires assertion | $p_{24}$ | $p_{10} \mid\!\sim p_{24}$ |
| — | Individuation $\Rightarrow$ Expanded | (from #11 follow-up) | Expanded Terms add expressiveness | $p_{23}$ | $p_{18} \mid\!\sim p_{23}$ |

- Read $p_2 \mid\!\sim p_{18}$ as: one cannot commit to $p_2$ (Entity is a thing with fixed aspects) while denying $p_{18}$ (fixed aspects are context-relative; change is modeled by deriving new Entities). Each accepted tension is one such defeasible rule
- Rationale codes are requirements in Moreau et al.'s retrospective analysis of the working group's record; six of the seven challenges land on one

# Verification with an automated reasoner

- We ran pyNMMS, our open-source reasoner, over the 19 propositions and 9 implications, asking *does Y follow from X?*: the analogue of an entailment check or a SPARQL `ASK`
- All 9 accepted implications are derivable; no proposition is both asserted and denied; the seven challenge chains are pairwise independent (34 pairs). And each structural property matches a documented design decision:

| What the reasoner found | Design decision (Moreau et al., 2015) |
|---|---|
| $p_2 \mid\!\sim p_{18}$ and $p_{18} \mid\!\sim p_{23}$ hold, but $p_2 \mid\!\sim p_{23}$ does not: inferences do not chain (nontransitive) | EZ3: minimal core, separate extensions |
| $p_2 \mid\!\sim p_{18}$ holds, but $p_2, p_{23} \mid\!\sim p_{18}$ does not: adding $p_{23}$ defeats it (nonmonotonic) | EZ1: "scruffy" to "proper" provenance |
| $p_7 \mid\!\sim p_{28}$ holds, but composes with no other chain | VI5, ISSUE-612: non-transitive derivation |
| No Entity commitment entails an Activity or Agent one (8 queries) | VI1, GE1: three independent views |
| The retracted $p_{20}$ leaves no consequence behind | GE3: retraction leaves no trace |

# RDF serialization with provenance

```
:i1 a mb:MaterialImplication, prov:Entity ;
    rdfs:label "p2 |~ p18" ;
    mb:antecedent :p2 ; mb:consequent :p18 ;
    prov:wasGeneratedBy :move11 .
:move11 a mb:DialogueMove, prov:Activity ;
    rdfs:label "Challenge #11 resolved by refinement" ;
    prov:wasAssociatedWith :respondent, :opponent ;
    prov:endedAtTime
        "2026-02-09T00:00:00Z"^^xsd:dateTime .
```

One material implication and the dialogue move that generated it, from `material-base.ttl`

- Propositions are resources; each material implication is reified and `prov:wasGeneratedBy` the dialogue move that accepted it, which links to the GitHub issues
- Validates against PROV-CONSTRAINTS (ProvValidator); the report is archived with the data
- The vocabulary the case study examined describes the dialectic that examined it

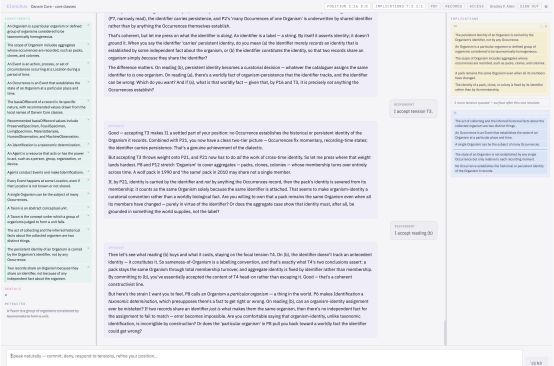
# Limitations

- The base serializes to RDF, but its consequence relation **cannot be delegated to RDFS/OWL entailment without loss**: monotonic, transitive entailment would license exactly the sequents the base rejects
- The case study is an **existence proof**, not an evaluation
- The current architecture pairs one expert with an LLM; **multiple experts and LLMs collaborating on a knowledge graph are not supported**; reconciling divergent positions across respondents is a open question
- Systematic oracle bias in tension identification is a possibility; **false negatives are invisible to both parties**

# Current work: full support for RDF knowledge graph construction

- **An implication-space semantics for RDF** (Allen, 2026), after Hlobil (2026): simple, RDFS and rule-defined OWL 2 RL entailment, blank nodes and inconsistency included, recovered *exactly* as the **monotone corner** of the base's semantics, by the same homophonic reading: each triple's content is its own bearer's role. Lean-verified; matches `owlrl` on ∼8M graph pairs
- So regime closure and *defeasible* commitments (denials, defaults with exceptions, curator overrides) sit in **one consequence relation**, with no change to RDF, RDFS, OWL 2 RL or SHACL: a step toward the "semantics for shared inference" of Hitzler et al. (2010)
- **Scalable implementation**: pyNMMS integrated with Oxigraph runs NMMS proof search against the regime's closure at $10^7 - 10^8$ triples, cost flat in graph size

# Current work: user studies to evaluate utility and usability




The elenchus-server user interface

- **A funded, pre-registered study**: thirty biodiversity scientists each specify a matched pair of domains under two counterbalanced conditions, free-form LLM conversation and Elenchus, model held fixed; a blinded panel of five expert judges rates the outputs on established ontology-quality criteria
- **Supporting infrastructure**: elenchus-server, a second, independent implementation; model-agnostic, multi-user, typed speech-act API, derivability queries, persisted in DuckDB

# Summary

- Elenchus is a proof-of-concept that shows how knowledge engineering can be accomplished through expert-AI natural language dialogue
- The expert is never asked to translate their knowledge into a target formalism: the dialogue elicits the consequence relation itself
- In an ontology engineering context, the deliverable is not only the ontology but **a defensible record of why it is the way it is**
- The LLM's unreliability is contained by the dialectic, not mitigated after the fact through a human-in-the-loop

# References i


Allen, Bradley P. (2026). **"Implication-Space Semantics for RDF".** Preprint; arXiv number pending. Lean 4 formalization and machine checks at `https://github.com/bradleypallen/iss-rdf-lean` and `https://github.com/bradleypallen/iss-rdf-checks`.

Allen, Bradley P., Lise Stork, and Paul Groth (2023). **"Knowledge Engineering Using Large Language Models".** In: *Transactions on Graph Data and Knowledge* 1.1, 3:1–3:19. doi: 10.4230/TGDK.1.1.3.

Dutilh Novaes, Catarina (2020). ***The dialogical roots of deduction: Historical, cognitive, and philosophical perspectives on reasoning*.** Cambridge University Press. doi: 10.1017/9781108800792.

Garijo, Daniel et al. (2024). **"LLMs for ontology engineering: a landscape of tasks and benchmarking challenges".** In: *Proceedings of the Special Session on Harmonising Generative AI and Semantic Web Technologies (HGAIS 2024) co-located with the 23rd International Semantic Web Conference (ISWC 2024)*. Vol. 3953. CEUR Workshop Proceedings. CEUR-WS.org, pp. 1–7. url: `https://ceur-ws.org/Vol-3953/364.pdf`.

# References ii


Hitzler, Pascal and Frank van Harmelen (2010). **"A reasonable Semantic Web".** In: *Semantic Web* 1.1–2, pp. 39–44. doi: 10.3233/SW-2010-0010.

Hlobil, Ulf (2026). **"First-order implication-space semantics".** In: *Journal of Philosophical Logic* 55.3, pp. 529–554. doi: 10.1007/s10992-026-09841-x.

Hlobil, Ulf and Robert B Brandom (2025). ***Reasons for logic, logic for reasons: Pragmatics, semantics, and conceptual roles*.** Routledge.

Huang, Lei et al. (2023). **"A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions".** In: *arXiv preprint arXiv:2311.05232*.

Koutsiana, Elisavet et al. (2026). **"Knowledge prompting: How knowledge engineers use generative AI".** In: *Journal of Web Semantics* 88, p. 100873. doi: 10.1016/j.websem.2025.100873.

Lebo, Timothy, Satya Sahoo, and Deborah McGuinness (2013). ***PROV-O: The PROV Ontology*.** W3C Recommendation. W3C. url: https://www.w3.org/TR/prov-o/.

# References iii


Moreau, Luc et al. (2015). **"The rationale of PROV".** In: *Journal of Web Semantics* 35, pp. 235–257.

Restall, Greg (2005). **"Multiple conclusions".** In: *Logic, methodology and philosophy of science: Proceedings of the twelfth international congress*. Kings College Publications, pp. 189–205.

Zhang, Bohui et al. (2025). **"OntoChat: A Framework for Conversational Ontology Engineering Using Language Models".** In: *The Semantic Web: ESWC 2024 Satellite Events*. Vol. 15344. Lecture Notes in Computer Science. Cham: Springer, pp. 102–121. doi: 10.1007/978-3-031-78952-6_10.

Zhao, Yihang et al. (2024). **"Improving ontology requirements engineering with OntoChat and participatory prompting".** In: *Proceedings of the AAAI Symposium Series*. Vol. 4:1, pp. 253–257.